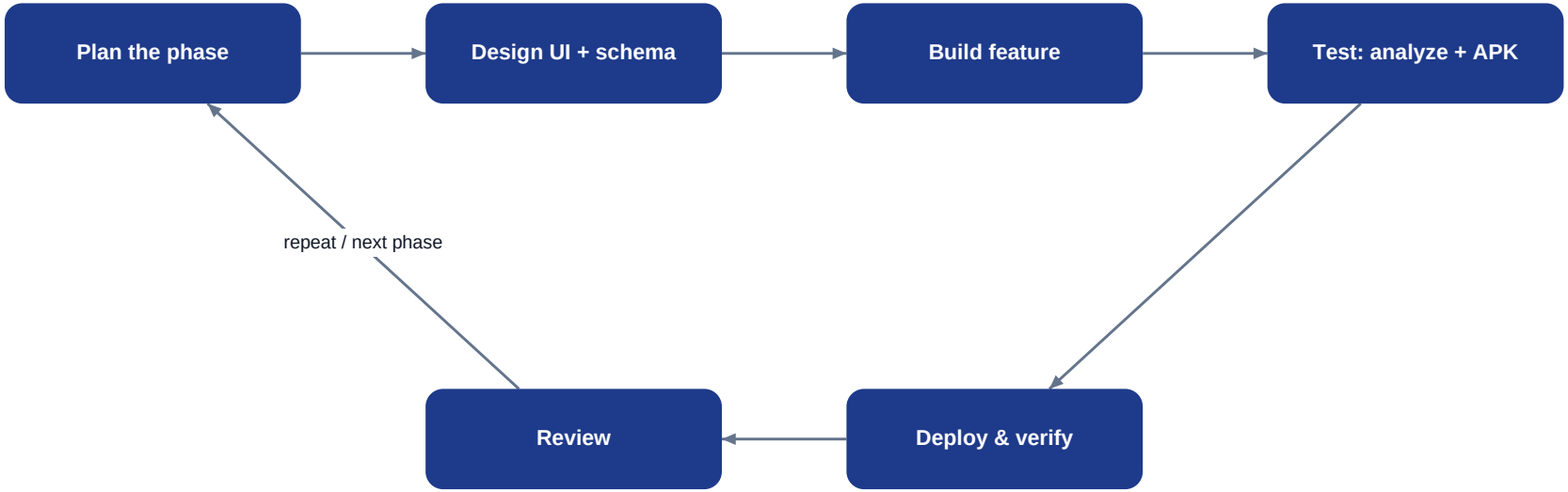
SkillX

System Design Diagrams

Chapter 6 | 14 Figures | Flutter + Firebase

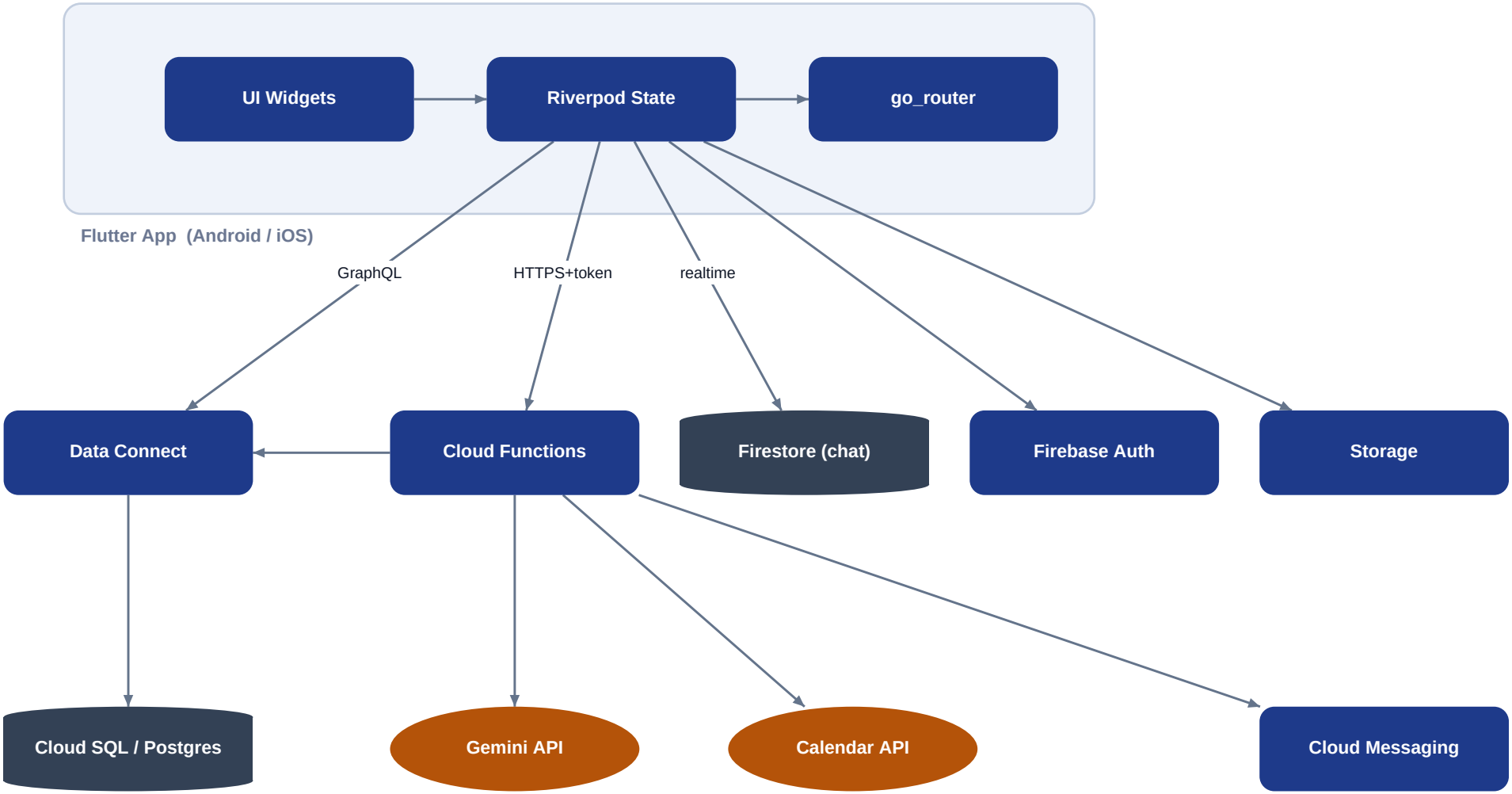
Peer-to-peer skill-swapping mobile application

FIG 1 Software Methodology Used



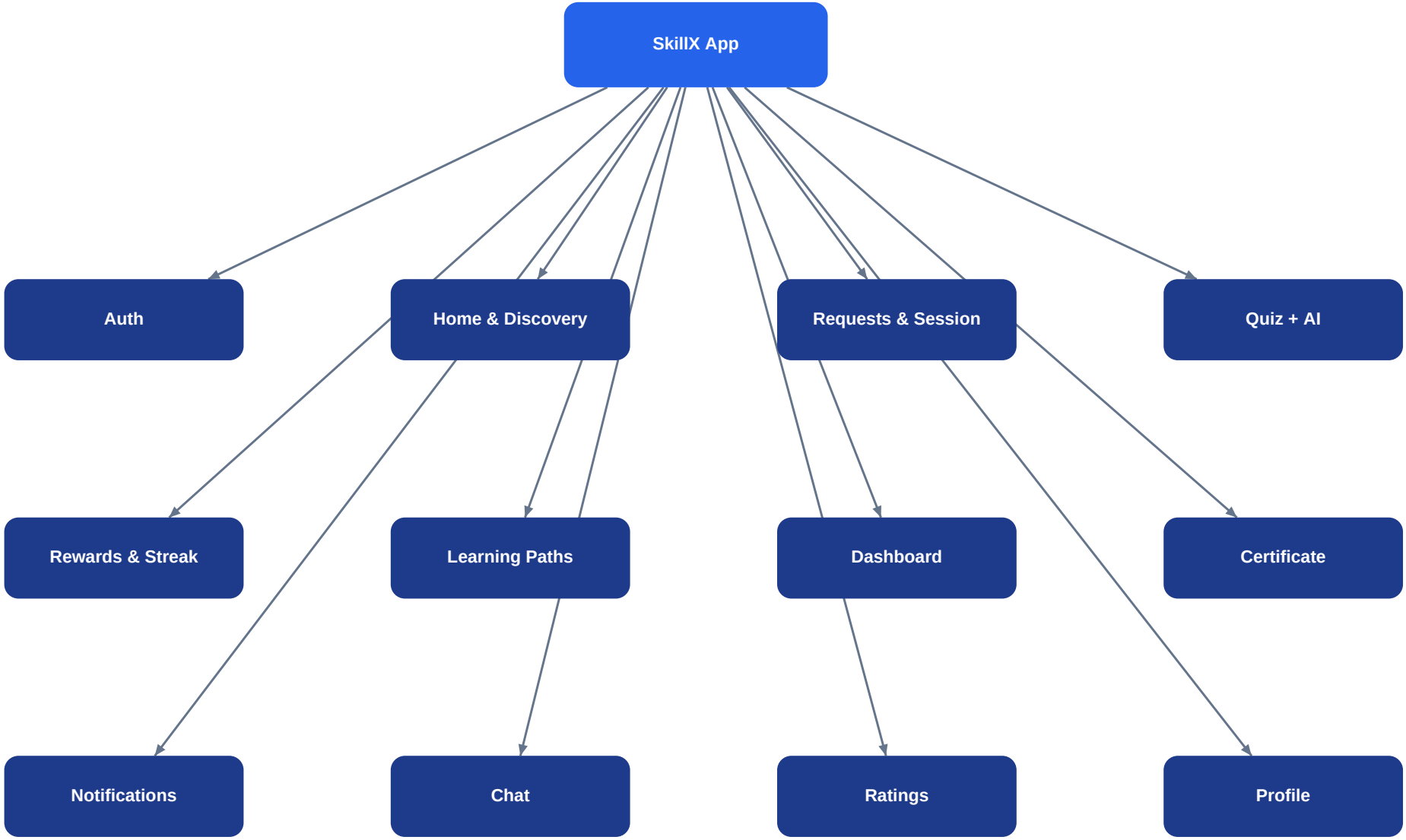
Agile / iterative model. SkillX was built in 13 short phases; each phase repeats this build-test-deploy loop.

FIG 2 System Architecture



Three-tier hybrid backend: the app talks to Firebase; structured data goes to Cloud SQL via Data Connect, chat to Firestore, and privileged logic to Cloud Functions, which call Gemini and Calendar.

FIG 3 Module Diagram

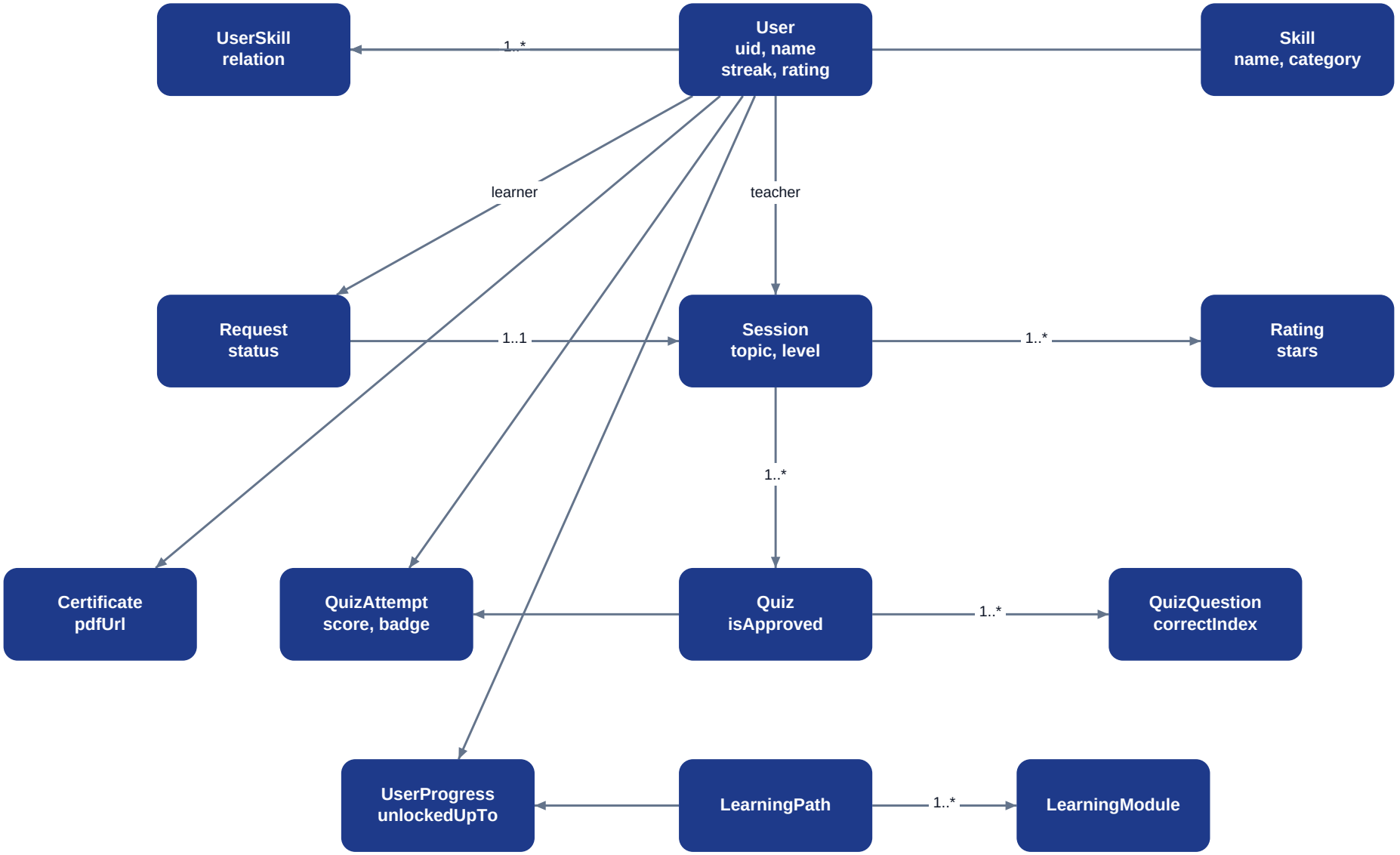


The app is split into 12 self-contained feature modules, each with its own domain / data / state / UI layers.

FIG 4 Use Case Diagram

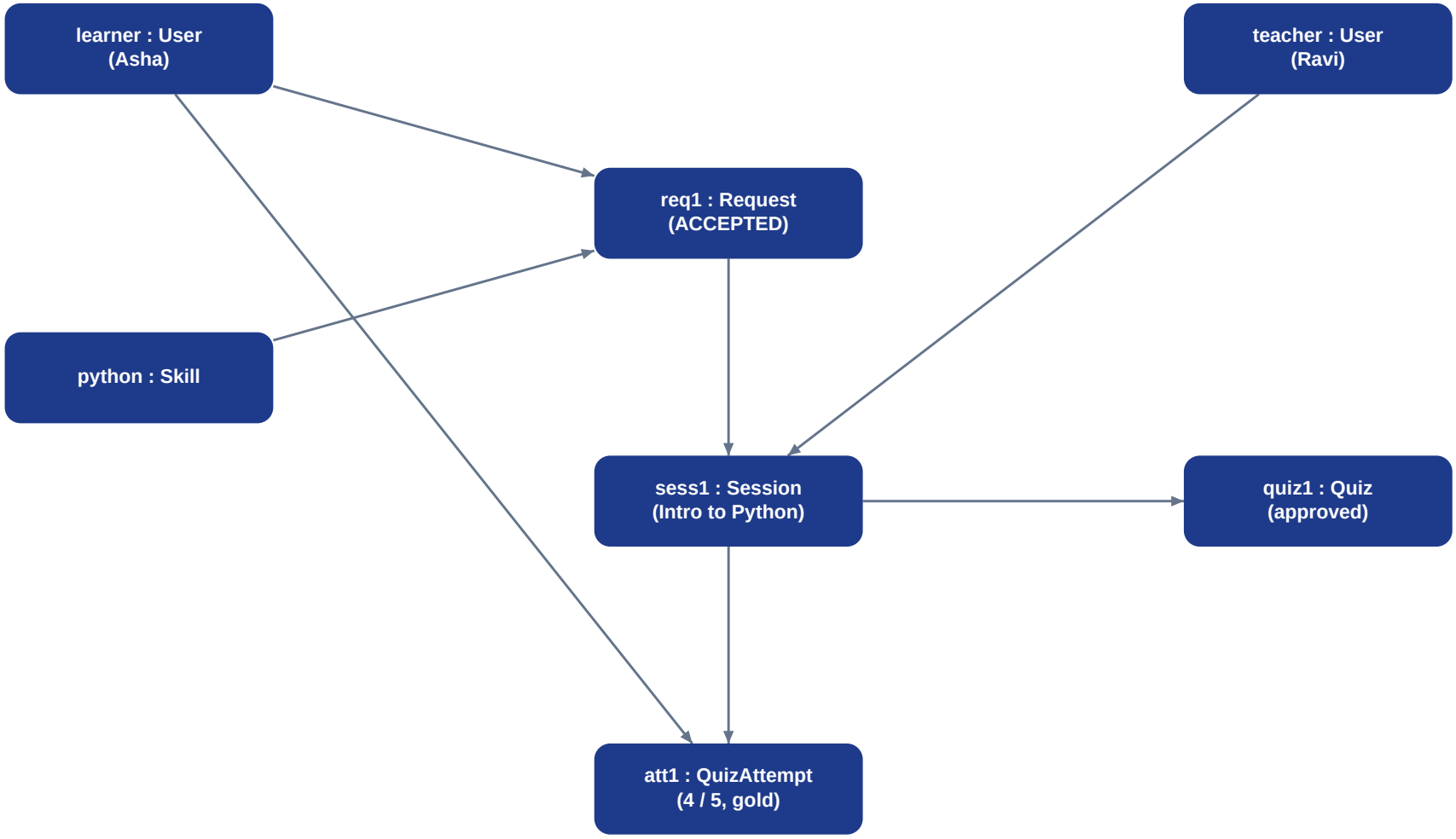
Actors and use cases. Every user is both a Learner and a Teacher; the lines show which actions each role performs.

FIG 5 Class Diagram



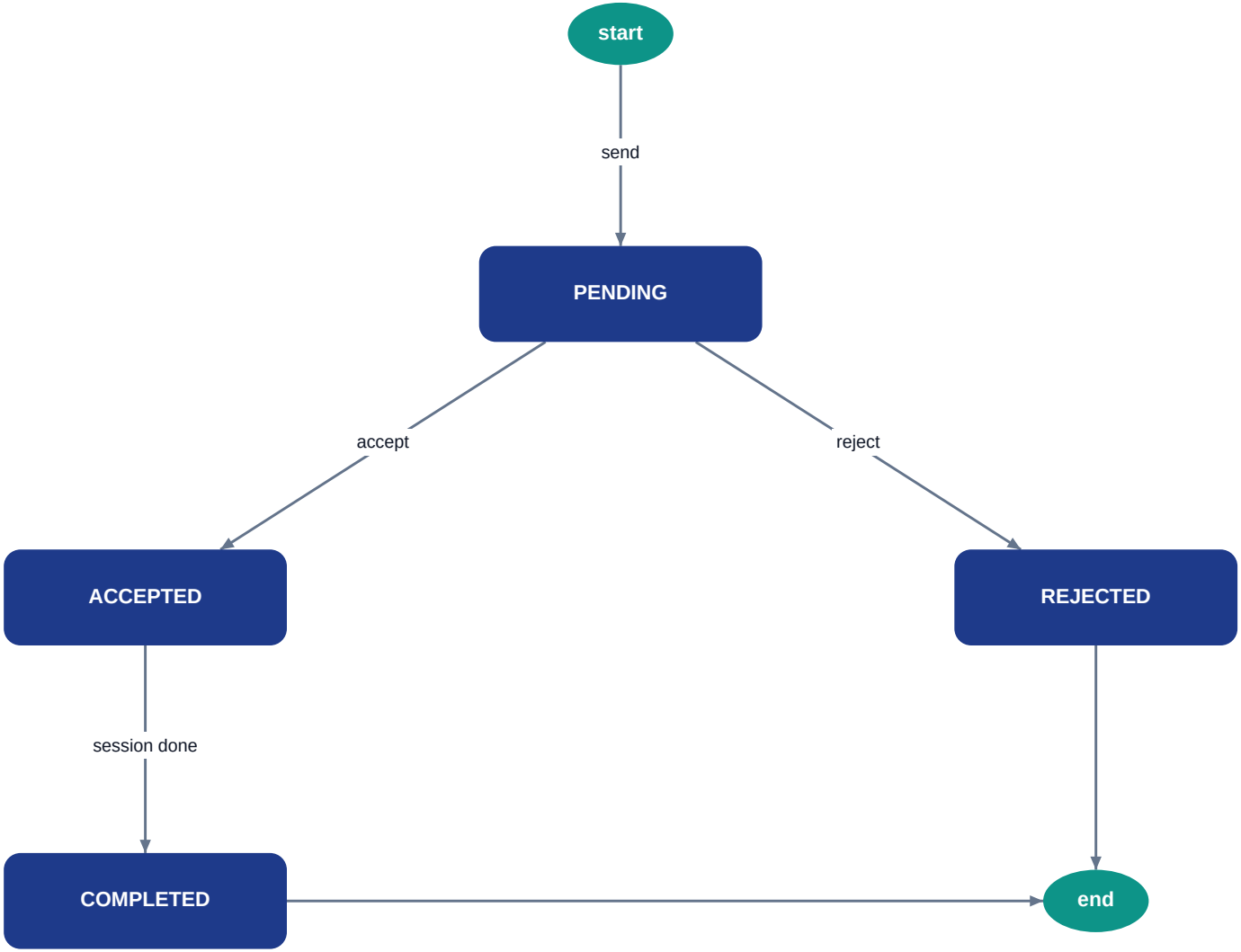
The main domain classes with key attributes and their 1-to-many relationships. These map directly to the Cloud SQL tables.

FIG 6 Object Diagram



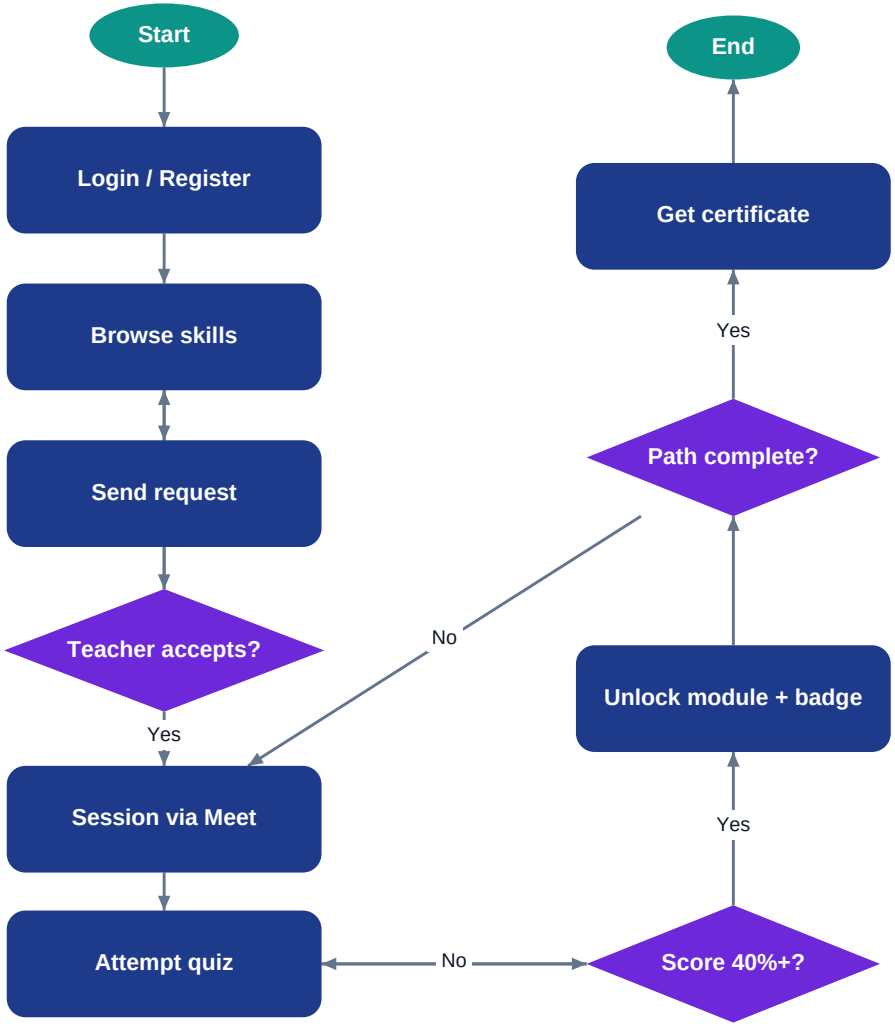
A snapshot of live objects: learner Asha requests Python from Ravi; it becomes a session, then a quiz, then a graded attempt.

FIG 7 State Diagram



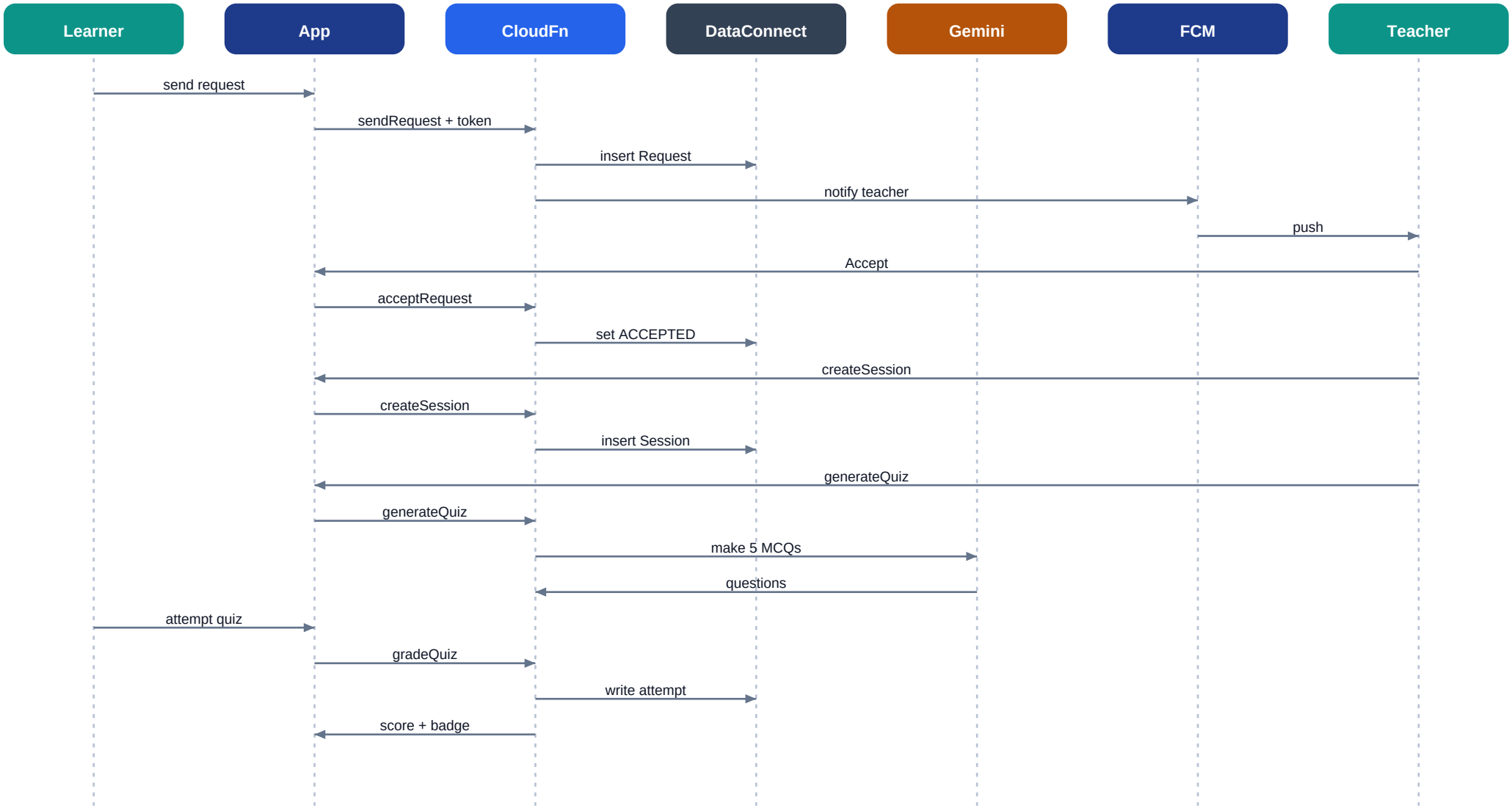
The states a learning request moves through and the events that cause each transition.

FIG 8 Activity Diagram



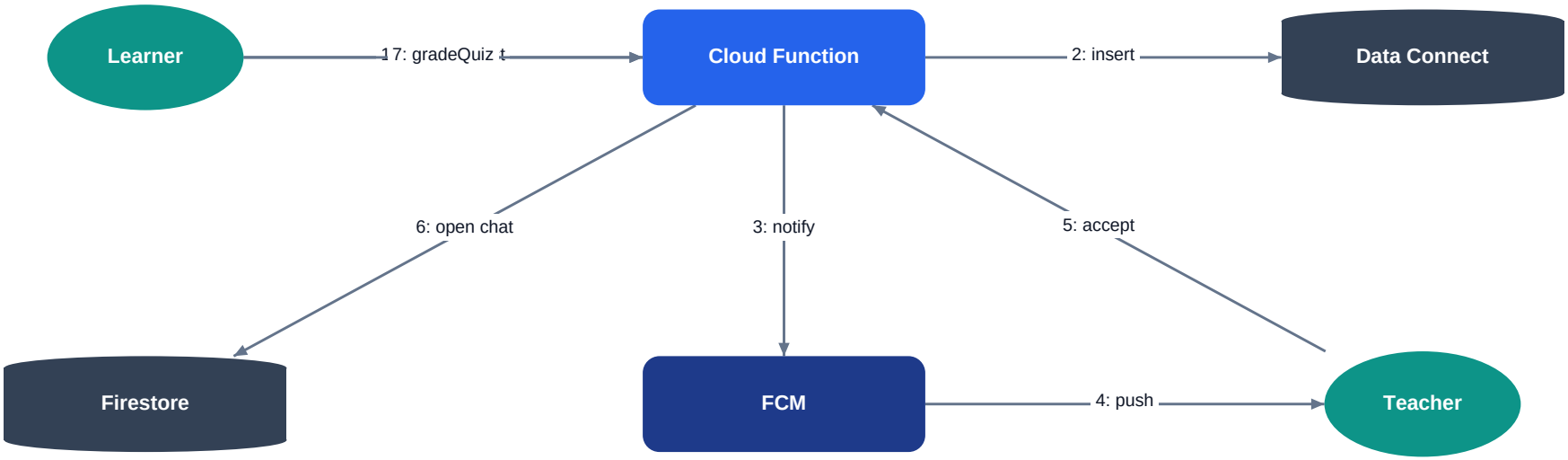
The end-to-end learn-a-skill workflow with its decision points (accept, pass mark, path completion).

FIG 9 Sequence Diagram



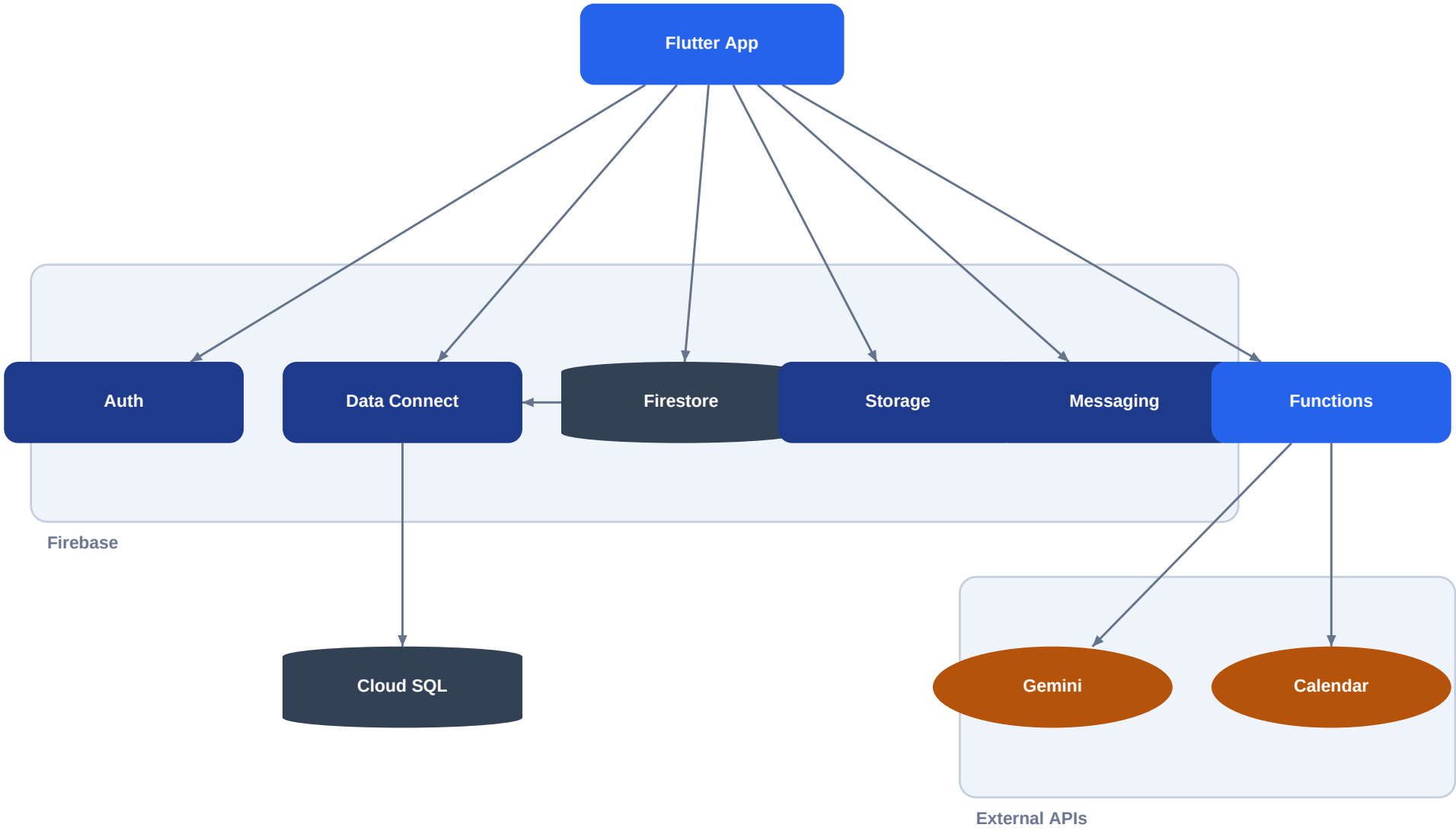
Time-ordered interactions for the core flow: request, accept, session, AI quiz, and server-side grading.

FIG 10 Collaboration Diagram



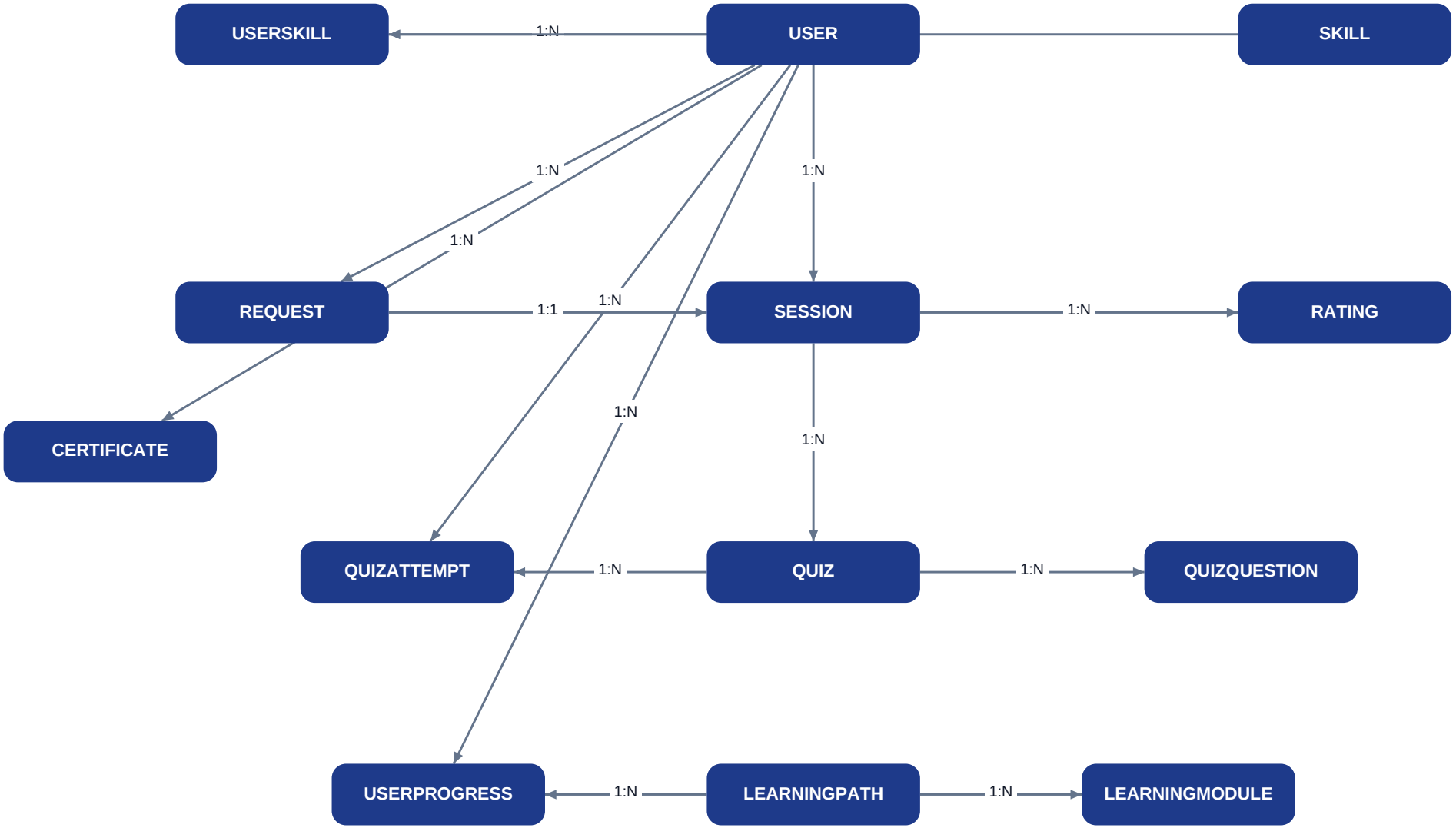
The same interactions as the sequence diagram, shown as numbered messages between collaborating objects.

FIG 11 Component Diagram



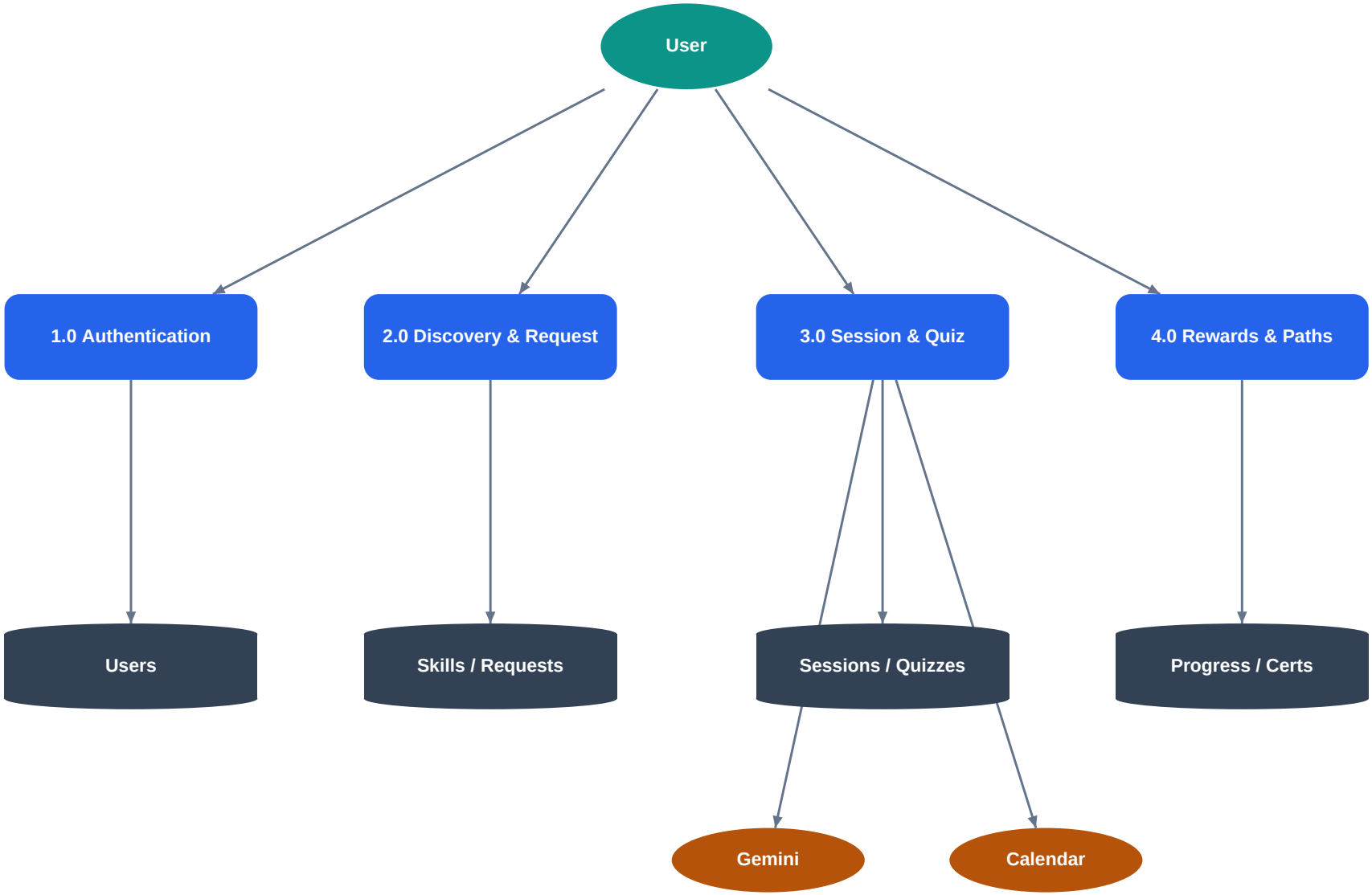
The deployable components and their dependencies - the physical building blocks of the running system.

FIG 12 E-R Diagram



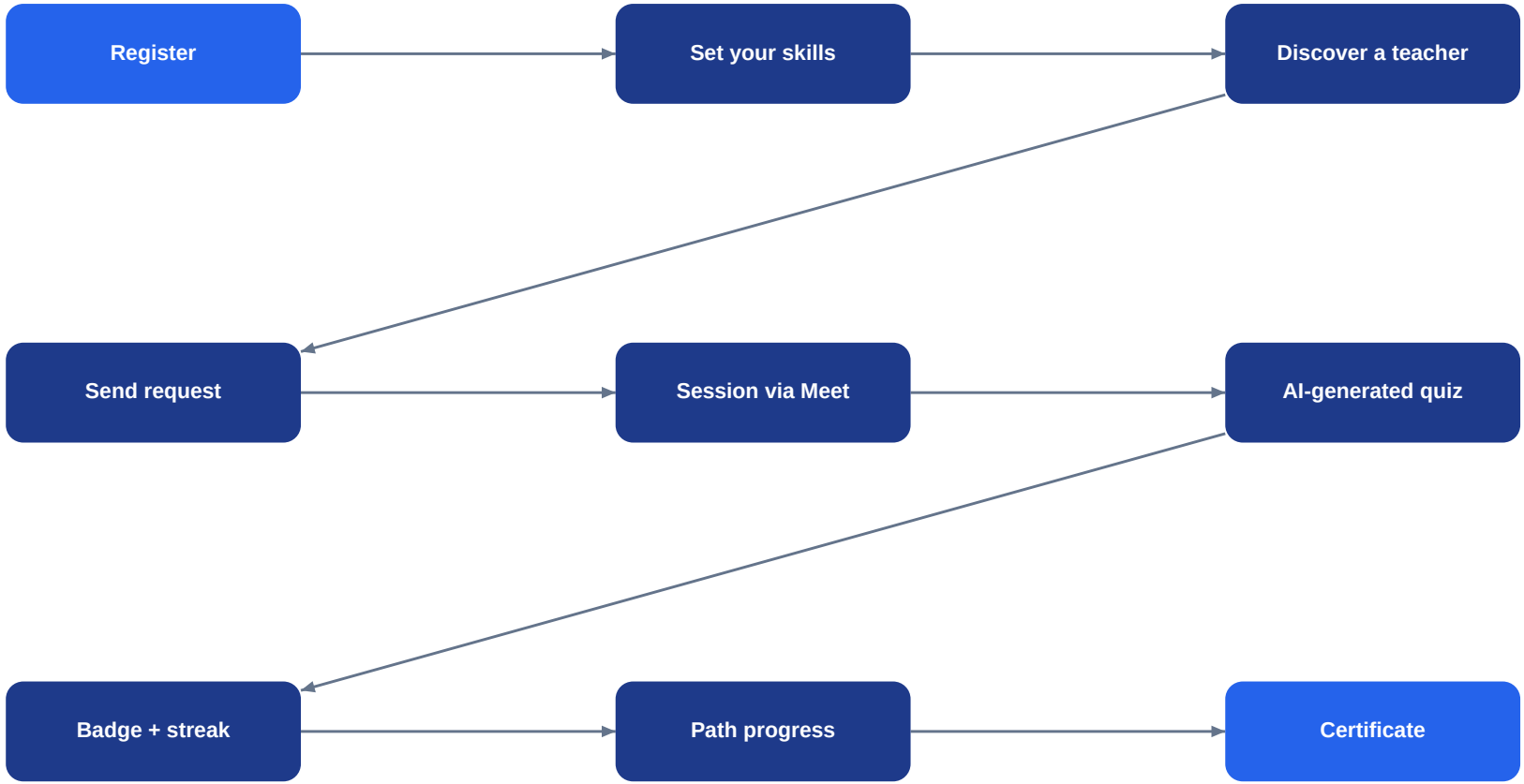
The entity-relationship model of the Cloud SQL database - the tables and how they relate (1:N = one to many).

FIG 13 Data Flow Diagram (Level 1)



How data flows between the user, the four main processes, the data stores, and the external services.

FIG 14 Project Flow Diagram



The full user journey from sign-up to certificate - a one-line summary of the whole product.